proposal

A Practical Evaluation of Oversmoothing and Oversquashing in Graph Neural Networks

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Abstract: Message-passing graph neural networks (MPNNs) are often reported to perform best at small depths, while deeper variants may deteriorate in accuracy or become difficult to train. The two most commonly invoked explanations are oversmoothing, meaning that node representations become more alike as depth increases and oversquashing, meaning that information from an exponentially growing receptive field has to be squeezed into a fixed-size vector, especially across topological bottlenecks. These definitions have been refined in recent research, which has also suggested mitigation strategies, but this research also questions the universal applicability of such explanations to performance drops in practical benchmarks. In particular, one approach attributes oversquashing to graph topology through commute time and effective resistance and highlights that width can help, while depth eventually encounters the vanishing gradient problem [DGGB$^+$23]. A different line of work unifies oversmoothing definitions and emphasizes that reducing oversmoothing is not enough to build strong deep GNNs [RBM23]. Newer position papers argue that widely repeated "universal" claims about oversmoothing and oversquashing have been overblown or misunderstood and that practical relevance depends on the task and how label-relevant information is distributed [ARE26, KDL26].

This bachelor thesis aims to conduct an appropriately scoped experimental study that compares theoretical expectations about oversmoothing and oversquashing with empirical behavior on a limited set of synthetic long-range transfer tasks (Ring, CrossedRing, CliquePath) and a small set of real-world node-classification datasets (a realistic subset drawn from Cora, CiteSeer, PubMed, Roman-Empire, Amazon Photo, Amazon Computers). The study will compare a small number of architectures two standard baselines plus one oversmoothing mitigation and one oversquashing mitigation approach and evaluate how performance, stability across depths, and simple diagnostic indicators (e.g. Dirichlet energy for oversmoothing) change as depth, width and topology vary.

The expected outcome is not a new theorem or a new architecture, but a grounded interpretation of when oversmoothing and oversquashing meaningfully help explain performance degradation and when other factors (dataset structure, graph topology, evaluation protocol) appear more relevant. The thesis goal will be to come up with a reproducible set of experiments and cautious, evidence-based conclusions.

1 Introduction and motivation

MPNNs iteratively update node representations by aggregating information from local neighborhoods, so information from nodes at graph distance $\mathbf{d}$ can only influence each other after roughly $\mathbf{d}$ message-passing layers. This naturally makes depth as an effective tool to capture long-range dependencies; however, in many practical settings deep message-passing stacks often underperform or become unstable when compared to shallow ones. This mismatch is often attributed to oversmoothing and oversquashing, but the literature has evolved into a situation where (i) various definitions and metrics coexist, (ii) mitigation strategies that target different failure cases and (iii) research papers that challenge general assumptions that these phenomena are the universal reason behind the depth-induced degradation across benchmarks [RBM23, KDL26]. In the case of oversquashing, according to recent theoretical advances, the challenge is not just about the "distance" but also about topology: information flow is hindered by bottlenecks and is strongly related to graph quantities such as commute time and effective resistance. Increasing width can improve sensitivity to distant nodes, while simply adding more layers does not solve this problem in the long run and may eventually lead to vanishing-gradient issues [DGGB$^+$23]. Regarding oversmoothing, survey papers suggest an axiomatic definition based on the exponential convergence of node-similarity measures. In addition to this, they address the various approaches to measure oversmoothing, such as Dirichlet energy, as well as different mitigation strategies. At the same time, they stress that reducing oversmoothing alone is not enough to make deep GNNs work well [RBM23]. At the same time, recent position papers argue that many widely repeated assumptions blur the distinctions between oversmoothing, oversquashing, long-range behavior, and heterophily, and that carefully constructed counterexamples show why broad universal claims should be treated with caution [ARE26]. Moreover, another position paper states that, for most benchmark tasks, accuracy does not seem to correlate significantly with commonly used metrics for detecting oversmoothing and that many interventions aimed at oversquashing lead to only limited improvements. Instead, it suggests paying closer attention to whether a task actually depends on non-local information and to how this information is distributed or localized across the graph [KDL26].

This thesis is motivated by the practical value of clarifying these debates, it aims to systematically examine selected claims in a controlled yet manageable experimental setting and to identify which findings appear robust and which seem to depend on the dataset or experimental protocol.

2 Problem formulation and research questions

2.1 Problem statement

The central problem is that oversmoothing and oversquashing are widely used as explanations for why deeper message-passing GNNs may degrade in performance, but the strength

and universality of these explanations in practical settings is disputed. A practically oriented experimental thesis should therefore ask the following question: across a modest set of tasks and datasets, do oversmoothing and oversquashing consistently align with patterns of performance degradation, and do mitigation strategies lead to improvements that match theoretical expectations?

This thesis focuses on message-passing architectures rather than full graph transformers and keeps the scope of models and datasets deliberately small in order to enable an interpretable comparison of (i) depth sweeps, (ii) width sweeps and (iii) topological differences between synthetic controlled graphs and real benchmark graphs.

2.2 Research objectives

1. Build a small, reproducible benchmark suite that includes both controlled synthetic "graph transfer" tasks and standard real-world node classification tasks, enabling systematic depth/width/topology sweeps.
2. Empirically measure how performance changes with depth for baseline MPNNs, and test whether declines align with oversmoothing indicators (e.g. Dirichlet energy) and with longrange difficulty in synthetic tasks.
3. Compare baseline models with one oversmoothing mitigation method (e.g. PairNorm or GCNII) and one oversquashing mitigation method (e.g. a rewiring-based approach), assessing whether these interventions improve depth robustness in practice.
4. Interpret results critically in light of the literature, including the "standard" perspective that oversmoothing/oversquashing limit deep GNNs and the newer critique that these phenomena are often overstated or misdiagnosed.
5. Produce a coherent set of evidence-based conclusions while also acknowledging the main methodological limitations.

2.3 Research questions

The study will be guided by the following research questions

- **Depth-performance relationship:** Does increasing depth consistently worsen nodeclassification performance across chosen real-world datasets, or are there datasets where moderate depth is beneficial?
- **Oversmoothing diagnostics:** Do simple oversmoothing indicators (e.g.Dirichlet energy) correlate with the depth at which performance degrades, or is the correlation weak / inconsistent across datasets and architectures?

- **Oversquashing on controlled tasks:** On synthetic graph-transfer tasks (Ring, CrossedRing, CliquePath), how do distance , graph topology, and hidden dimension affect performance, and do observed patterns align with theoretical expectations that width can mitigate oversquashing while topology strongly shapes
- **Mitigation effectiveness:** Do selected mitigation methods provide meaningful and robust gains (across depths and random seeds), or only limited improvements confined to specific tasks/datasets?
- **Topology vs. architecture:** Is graph topology (e.g. bottlenecks, neighborhood growth) more predictive of long-range difficulty than architectural depth alone, especially in the synthetic setting designed to isolate oversquashing?
- **Explanatory adequacy:** Are some practical performance drops better explained by task structure and dataset properties than by general claims about oversmoothing or oversquashing?

3 State of research

3.1 Oversmoothing

Oversmoothing is commonly described as the tendency of node embeddings to become increasingly similar as additional message-passing layers are applied, resulting in collapse into a low-dimensional subspace or even near-constant signals. In the paper A Survey on Oversmoothing in Graph Neural Networks, Rusch et al. define oversmoothing as the exponential convergence of the similarity between nodes as the network's depth increases, aiming to simplify the understanding and measurement of the phenomenon [RBM23].

Within this line of work, Dirichlet energy, together with closely related measures of graph smoothness, is widely used to capture how similar the representations of neighboring nodes are.It is introduced as a proper similarity metric in accordance with the suggested set of axioms [RBM23]. The survey also highlights an important practical point: even if oversmoothing is reduced, in the sense that embeddings no longer collapse, this would not suffice for creating expressive deep GNNs that perform well in any task setting. In other words, preventing representation collapse does not automatically mean that deeper models will learn useful discriminative features [RBM23].

Various methods have been proposed to mitigate this issue. One example is PairNorm, a representative approach that adds a normalization layer to prevent embeddings from becoming too similar and can be integrated easily without modifying the overall architecture [ZA20]. GCNII is an example of an architectural modification that enables deep GCN-style architectures by including initial residual connections and identity mapppings and is proposed as a way of solving the problem of oversmoothing [CWH$^+$20].

Nevertheless, recent criticisms have found that there may be some exaggerations regarding the issue of oversmoothing. In their ICLR 2026 paper the authors argues that different

oversmoothing metrics can lead to conflicting conclusions and that oversmoothing should not be treated as a simple or universal phenomenon across architectures and training settings. Instead, the paper cautions against broad generalizations and calls for more precise research questions [ARE26].

Similarly, Kormann et al. argue, based on experiments on standard benchmarks, that accuracy and common oversmoothing indicators are often only weakly related. They further show that the optimal model depth may still remain low even when mitigation methods are used, which suggests that oversmoothing is not always the main practical bottleneck [KDL26].

3.2 Oversquashing

Oversquashing refers to the difficulty of passing and preserving information from distant parts of a graph when many messages have to be aggregated into fixed-size node representations. From the bottleneck perspective, this problem can be understood as a form of information compression along long paths, where exponentially growing information is forced into bounded vectors. This is often linked to failures on tasks that depend on long-range interactions [AY21].

Topping et al. give a geometric-topological view using curvature-based rewiring. They argue that negatively curved edges are associated with oversquashing and use this idea to motivate curvature-based rewiring as a way to reduce structural bottlenecks [TDGC$^+$22].

The ICML 2023 paper "On Over-Squashing in Message Passing Neural Networks: The Impact of Width, Depth, and Topology" extends the theoretical understanding of oversquashing and and puts forth several claims that are directly relevant for the design of a practical study. To begin with, the research suggests that model widening can be used to combat oversquashing even though this might lead to increased sensitivity. Second, it shows that simply increasing depth does not solve oversquashing in the long run and may instead lead to regimes dominated by vanishing gradients. Third, the research highlights the importance of graph topology by linking oversquashing to quantities such as commute time and access time between nodes, thereby offering a common perspective for understanding rewiring-based approaches [DGGB$^+$23].

Notably, the accompanying code repository of that work provides a concrete empirical setup for validation through graph-transfer tasks on the synthetic topologies Ring, CrossedRing, and CliquePath. It also emphasizes that the meaning of "long-range" depends strongly on the underlying topology. In addition, the paper reports experiments based on Jacobian-related difficulty indicators and effective-resistance-related measures to analyze topological effects [DGGB$^+$23].

3.3 Long-range benchmarks and why synthetic tasks remain useful

Long-range behavior is difficult to diagnose on standard benchmarks because many node-classification datasets may not require truly non-local information for good performance. The Long Range Graph Benchmark (LRGB) was proposed specifically to evaluate models on tasks that arguably demand longrange interaction reasoning, and it benchmarks message-passing GNNs against graph transformer approaches [DRG+22].

Small synthetic tasks remain valuable for isolating oversquashing and oversmoothing mechanisms because they allow controlled manipulation of distance, depth, width, and topology while reducing confounding effects from real-world dataset quirks.

3.4 Relevant mitigation approaches

For oversmoothing, PairNorm and GCNII are representative interventions with publicly available descriptions and code, and they are commonly used in discussions of depth-robust message passing [ZA20, CWH+20].

For oversquashing, many methods can be viewed as forms of graph rewiring or alternative propagation mechanisms aiming to reduce bottlenecks. DiffWire proposes a differentiable and inductive approach grounded in commute time learning and spectral gap considerations, explicitly aiming to address limitations including oversquashing [ARBEO22]. Expander Graph Propagation (EGP) proposes using expander graphs to support global communication without dense all-to-all attention, framing this as a way to avoid bottlenecks and oversquashing while keeping complexity manageable [DLV22].

At the same time, the recent critique literature emphasizes that mitigation methods should not be assumed to provide consistent gains on standard benchmarks, and that empirical evaluation should be cautious, well-controlled, and explicit about what is being tested [ARE26, KDL26].

4 Research gap and contributions

4.1 Research gap

A noticeable gap is the lack of small, controlled studies that jointly examine oversmoothing and oversquashing in the same experimental protocol, across both (i) synthetic tasks explicitly constructed to stress long-range propagation and (ii) standard node classification benchmarks that may or may not require long-range information. The core oversquashing paper provides a strong theoretical framework and some synthetic validation tooling, but it is not primarily framed as a "cross-benchmark practical relevance" study [DGGB+23]. The oversmoothing survey provides definitions and empirical illustrations across graphs and mitigation methods, but it does not settle how diagnostic measures relate to predictive

accuracy in a way that holds uniformly across datasets [RBM23].

More importantly for motivation, recent papers explicitly point out that the field contains many overly broad beliefs, metric ambiguities, and conceptual conflations; they argue that some commonly accepted statements are not generally true and that practical relevance depends on task structure and dataset properties [ARE26, KDL26] . This creates an ideal niche for a thesis in which a careful experimental design can clarify which claims appear robust under modest replication and which appear fragile or context-dependent.

4.2 scientific contribution

The thesis's scientific contribution will be modest but concrete: (i) a reproducible experimental suite linking theory-motivated diagnostics (e.g., Dirichlet energy, depth/width sweeps, topology statistics) to practical performance, (ii) a structured comparison of "standard view" vs. "skeptical view" claims in a controlled scope and (iii) a careful discussion of methodological pitfalls (metric choice, split choice, and confounding factors) grounded in recent critique papers [ARE26, KDL26].

5 Methodology and experimental design

5.1 Proposed methodology

The thesis will implement a small suite of MPNN models and mitigation variants using a widely used graph learning framework (e.g., PyTorch Geometric) and run systematic experiments with controlled sweeps over depth and width. The focus will be on interpretability and comparability, not on maximizing leaderboard scores. The methodology is designed around three principles:

1. **Controlled variation:** vary one factor at a time (depth, width, topology, mitigation) while keeping training protocol and splits fixed.
2. **Multiple signals:** evaluate both predictive performance and diagnostic measures (e.g., oversmoothing indicators), and analyze correlations rather than assuming causation.
3. **Reproducibility:** fixed random seeds, documented hyperparameters, and transparent reporting of runtime and training stability.

6 Synthetic experiments

The synthetic component is based on a graph transfer setup in which a graph is constructed with a source node and a target node placed at distance $\mathbf{d}$. The target node is assigned a unique one-hot feature, while all remaining nodes receive the same constant feature vector. The task for the model is to propagate the target signal through the graph so that it is reflected in the source nodes representation. As $\mathbf{d}$ increases, performance is typically expected to decline, making this setup a useful way to study how effectively information can travel across the graph [DGGB$^+$23].

Three synthetic topologies will be used: Ring, CrossedRing, and CliquePath. These graph types form the basis of the long-range dependency validation setup, with scripts available for varying both the node distance and the hidden dimension. The repository also notes that the topology changes the difficulty ordering [DGGB$^+$23].

6.1 Depth and width sweeps

For each topology, the distance $\mathbf{d}$ will be swept over a modest range. Two sweep types will be run:

1. **Width sweep:** Fix depth equal to distance and evaluate whether increasing hidden dimension improves transfer accuracy, reflecting the theoretical expectation that width can mitigate oversquashing [DGGB$^+$23].
2. **Depth sweep:** Fix $\mathbf{d}$ and vary depth around and beyond the distance scale, to observe whether adding layers helps, saturates, or destabilizes training, this aligns with the core papers claim that depth regimes may shift from oversquashing-dominated to vanishing-gradient dominated behavior [DGGB$^+$23].

7 Real-world experiments

7.1 Dataset selection strategy

The thesis will include a small, realistic subset of standard node-classification datasets rather than an exhaustive benchmark sweep. The guiding idea is to cover a few qualitatively different regimes:

Homophilous citation networks: Cora, CiteSeer, and PubMed are classic citation-network datasets commonly used to evaluate GCN-style models. They contain sparse bag-of-words node features, and citation links are usually treated as undirected edges in standard experiments [KW17].

Heterophilous / structurally different dataset: Roman-Empire is a structurally different dataset based on a Wikipedia article, where nodes represent words and edges capture adjacency or dependency relations. It is commonly used in benchmarking settings that focus on heterophily [Dee26].

Dense co-purchase graphs: Amazon Photo and Amazon Computers are co-purchase graph datasets in which nodes represent products, edges indicate frequently co-bought items, features are bag-of-words review vectors, and labels correspond to product categories. They are widely used as mid-scale benchmarks in graph machine learning [PyT26].

7.1.1 Proposed real-world subset (might change)

- Cora (small citation graph, widely used baseline)
- PubMed (larger citation graph, different scale and sparsity)
- Roman-Empire (heterophilous / text-derived dependency structure)
- Amazon Photo (dense co-purchase structure, mid-scale)

8 Models

Baseline architectures:

GCN as the primary baseline for simplicity and comparability to much of the oversmoothing/oversquashing literature [KW17].

GAT (or alternatively GraphSAGE) as a second baseline to test whether attention-based or sampling-based aggregation changes depth sensitivity and long-range behavior [VCC+18, HYL17].

One oversmoothing mitigation method:

PairNorm as a plug-in layer for stabilizing representation diversity across depth, suitable for both GCN and GAT and relatively easy to add without redesigning the model [ZA20].

(An alternative is GCNII, proposed by Chen et al. [CWH+20])

One oversquashing mitigation method:

Curvature-based rewiring motivated by the claim that bottlenecks and negatively curved edges are responsible for oversquashing; this provides a conceptually direct test of a prominent oversquashing mitigation idea [TDGC+22].

(If curvature computation proves too complex for the scope of the bachelor thesis, an alternative is to use the commute-time/effective-resistance perspective proposed by Di Giovanni et al. [DGGB+23].)

9 Evaluation criteria and diagnostics

Performance and Computational Cost

The evaluation is based on the following criteria:

1. Accuracy
2. Additionally, macro-F1 where class imbalance may matter, reported consistently across runs.
3. wall-clock training time per epoch and total training time
4. memory use (approximate) and whether models become unstable at depth.

Oversmoothing indicator The primary oversmoothing diagnostic will be Dirichlet energy of embeddings across layers, following the oversmoothing survey discussion of Dirichlet-energy-based similarity measures [RBM23].

Oversquashing-related diagnostics

- performance vs. distance $\mathbf{d}$ on graph-transfer tasks [DGGB$^+$23]
- a topology summary for synthetic graphs, used descriptively rather than as a formal proof instrument [AY21, TDGC$^+$22]

10 Experimental Protocol and Analysis Plan

Training protocol

For each dataset, the model will be trained using a standard supervised objective. Hyperparameters will be chosen in a minimal and consistent way by using a fixed optimizer family, a small fixed learning-rate grid, and early stopping based on validation performance. Feature preprocessing will also remain fixed for each dataset in accordance with standard library defaults in order to avoid additional confounding factors. In addition, it will be documented carefully which models require special stabilization steps such as increased dropout or gradient clipping, since these interventions may themselves be related to vanishing-gradient and depth issues discussed in the oversquashing literature [1].

Comparison matrix For each dataset type, the main comparison will be:

1. Baseline GCN
2. Baseline GAT (or GraphSAGE not sure yet)
3. Baseline + PairNorm (oversmoothing mitigation)
4. Baseline + rewiring (oversquashing mitigation)

Analysis approach Analysis will be reported in three layers:

1. Performance curves: accuracy/macro-F1 vs. depth for each model/dataset; synthetic transfer accuracy vs. distance $\mathbf{d}$.
2. Diagnostic curves: oversmoothing indicator vs. depth, descriptive topology measures for synthetic graphs; optional resistance proxy if feasible.
3. Correlation / interpretation: for each dataset, examine whether drops in performance align temporally (in depth) with diagnostic changes.

11 chapter agenda (might change)

- Introduction: motivation, problem framing, and contributions.
- Background: message passing and depth in GNNs, definitions of oversmoothing and oversquashing; why diagnostics are needed.
- Related work: oversmoothing literature, oversquashing literature, and recent critique/position papers, positioning of this thesis.
- Methodology: datasets, models, implementation details, training protocol, and chosen diagnostics; reproducibility measures.
- Experiments on synthetic long-range tasks: results, depth/width/topology analysis, and interpretation.
- Experiments on real-world datasets: depth sweeps, mitigation comparisons, diagnostic vs. accuracy correlation analysis.
- Discussion: synthesis across synthetic vs. real-world findings, where theory aligns, where it does not, limitations and threats to validity.
- Conclusion: summary of findings and practical recommendations.

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